

SUPPLEMENTAL MATERIAL

A. Notations

TABLE S-I
DEFINITION OF NOTATIONS IN THE MILP AND CP MODELS

Notations	Definition
i, i'	Indices of jobs
j, j'	Indices of operations
w, w'	Indices of workstations
r, r'	Indices of multi-robot collaborative processing modes
NJ	Number of jobs
NO_i	Number of operations of job i
NO_i^{SCO}	Number of SCOs of job i
NW	Number of workstations
$NR_{i,j,w}$	Number of robots of operation $O_{i,j}$ on workstation w
SJ	Set of jobs, $SJ = \{1, 2, \dots, NJ\}$
SO_i	Set of operations of job i , $SO_i = \{1, 2, \dots, NO_i\}$
SW	Set of workstations, $SW = \{1, 2, \dots, NW\}$
$SEW_{i,j}$	Set of eligible workstations of operation $O_{i,j}$
$SR_{i,j,w}$	Set of robots of operation $O_{i,j}$ on workstation w , $SR_{i,j,w} = \{1, 2, \dots, NR_{i,j,w}\}$
$pt_{i,j,w,r}$	Processing time of operation $O_{i,j}$ on workstation w with processing mode r
Δ^{PE}	Processing energy consumption rate
Δ^{SE}	Standby energy consumption rate
M	A sufficiently large positive constant

B. Encoding & Decoding

A three-vector encoding scheme is adopted to represent a solution of EE-MRC-FJSPDS, as shown in Fig. S-1. Each solution consists of an OS vector, a WS vector, and a RS vector. The OS vector determines the processing priority of operations, while the WS and RS vectors determine the selected workstation and multi-robot collaborative processing mode for each operation, respectively. In the OS vector, SCOs are encoded by their corresponding job indices. For example, the repeated symbols 1, 2, and 3 represent the SCOs of Job 1, Job 2, and Job 3, respectively. When the same job index appears multiple times, its occurrence order determines the corresponding SCO. For instance, the three occurrences of symbol 1 correspond to O11, O12, and O13, respectively. Different from SCOs, each SFO is encoded by a unique symbol. In Fig. S-1, symbols 4, 5, 6, and 7 represent O1a, O1b, O2a, and O3a, respectively. The WS and RS vectors are encoded according to the operation order within each job. Specifically, the first segment corresponds to Job 1, the second segment corresponds to Job 2, and the third segment corresponds to Job 3. For each operation, the value in the WS vector indicates the selected workstation, and the value in the RS vector indicates the selected multi-robot collaborative processing mode. For example, in Fig. S-1, operation O11 is assigned to workstation 2 with processing mode 4, while operation O22 is assigned to workstation 1 with processing mode 1.

The decoding procedure is conducted from left to right according to the OS vector. When a job index is encountered, the next unscheduled SCO of the corresponding job is selected. When an SFO symbol is encountered, the corresponding SFO is directly selected. Then, its workstation and multi-robot collaborative processing mode are obtained from the WS and RS vectors, respectively. The operation is then processed on the selected workstation using the selected processing mode at the earliest feasible time. After all operations are decoded, the makespan and TEC are calculated according to the generated schedule.

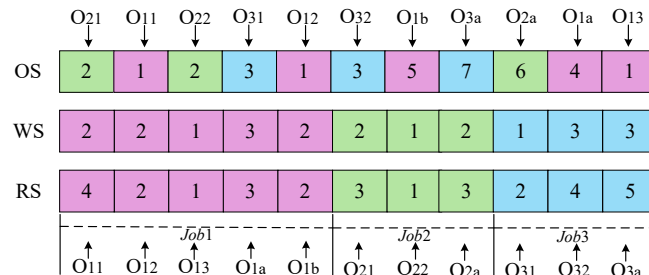


Fig. S-1. A solution of EE-MRC-FJSPDS.

C. Crossover & Mutation

The crossover and mutation operators used in task-internal evolution are described below. The same crossover operators are also employed for knowledge transfer.

Crossover: The crossover operators are illustrated in Fig. S-2. The precedence operation crossover (POX) operator is adopted for the OS vector. In POX, all jobs are first randomly divided into two subsets, denoted as $Jset1$ and $Jset2$. For each offspring, the operations belonging to one selected job subset are inherited from one parent and kept in their original positions. The remaining empty positions are then filled by scanning the other parent from left to right and sequentially inserting the operations belonging to the other job subset. The two-point crossover (TPX) operator is adopted for the WS and RS vectors. Two crossover points, denoted as a and b , are randomly generated. The segment between these two points is exchanged between the two parents, while the remaining positions are inherited from the original parent.

Mutation: The mutation operators are shown in Fig. S-3. For the OS vector, a swap operator is adopted. Two positions are randomly selected, and the corresponding genes are exchanged. For the WS and RS vectors, a reassignment operator is used. Specifically, one position is randomly selected, and its value is reassigned to another feasible value.

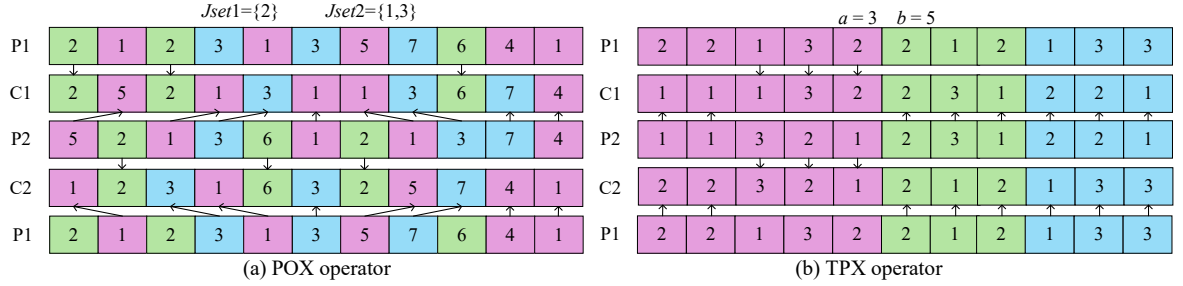


Fig. S-2. Crossover operators employed in KD-EMMTMA.

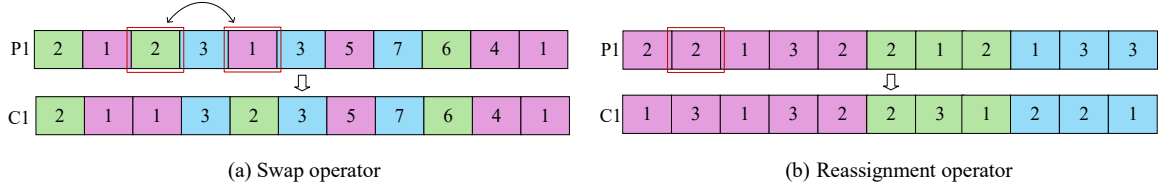


Fig. S-3. Mutation operators employed in KD-EMMTMA.

D. Domain Knowledge

Six properties are detailed below and illustrated in Fig. S-4.

Property 1: For a solution π , assume that operation $O_{i,j} \in \Pi$ is processed on workstation w . If there exists another eligible workstation $w' \in SEW_{i,j}$ such that changing the workstation of $O_{i,j}$ from w to w' makes the completion time of $O_{i,j}$ earlier without delaying its predecessors, then the makespan of the obtained solution π' can be decreased.

Proof: Since $O_{i,j}$ belongs to the critical path Π , its completion time directly affects the length of Π . Changing the workstation of $O_{i,j}$ may change its processing time and earliest start time. If the new workstation w' enables $O_{i,j}$ to complete earlier and does not delay its predecessors, a time gap can be generated on the original critical path. Then, the subsequent operations on Π can move forward. Therefore, the length of Π can be shortened, and $C_{\max}(\pi') < C_{\max}(\pi)$ may hold.

Property 2: For a solution π , assume that there exists a critical block $BlockP$ on the workstation w . If one operation $O_{i,j} \in BlockP$ is reassigned to another eligible workstation $w' \in SEW_{i,j}$, and the length of $BlockP$ is shortened without delaying the following critical operation, then the makespan of the obtained solution π' can be decreased.

Proof: If one operation in the block is moved to another eligible workstation, the workload concentration on workstation w can be relieved. When this reassignment shortens the length of $BlockP$ and does not delay the following critical operation, the subsequent critical operations can be shifted forward. Therefore, the length of the critical path decreases, and $C_{\max}(\pi') < C_{\max}(\pi)$ may hold.

Property 3: For a solution π , assume that an SFO $O_{i,j} \in \Pi$. If assigning $O_{i,j}$ a more suitable relative processing priority makes $O_{i,j}$ start earlier without delaying its preceding critical operations on Π , then the makespan of the obtained solution π' can be decreased.

Proof: If a critical SFO is assigned a more suitable processing priority, the waiting time before this SFO may be reduced. Once this SFO completes earlier, its succeeding operations on the critical path may also move forward. Therefore, the length of Π can be shortened, and $C_{\max}(\pi') < C_{\max}(\pi)$ may hold.

Property 4: For a solution π , assume that a critical block $BlockP$ exists on the critical path Π . If resequencing the operations in $BlockP$ makes one critical operation complete earlier and further reduces the waiting time of its succeeding critical operation without changing the workstation and multi-robot collaborative processing mode selections, then the obtained solution π' can dominate π .

Proof: The operations in $BlockP$ are consecutive operations processed on the same workstation. Therefore, the processing length of $BlockP$ itself does not change when workstation selections and collaborative processing modes remain unchanged. However, resequencing the operations in $BlockP$ may make a critical operation complete earlier, thereby allowing its succeeding critical operation to start earlier. In this case, the length of the critical path can be shortened, and $C_{\max}(\pi') < C_{\max}(\pi)$ may hold. Since workstation selections and collaborative processing modes remain unchanged, the processing energy consumption remains unchanged. Meanwhile, a smaller $C_{\max}$ reduces the standby energy consumption of workstations. Therefore, $TEC(\pi') < TEC(\pi)$ can be obtained. Since both $C_{\max}$ and TEC are improved, π' dominates π .

Property 5: For a solution π , assume that a segment of consecutive operations exists on the critical path Π . If adjusting the relative processing priorities of the operations in this segment reduces the total waiting time between adjacent critical operations without delaying the preceding critical operations on Π , then the makespan of the obtained solution π' can be decreased.

Proof: A segment of consecutive operations on Π directly contributes to the makespan. The waiting time between adjacent critical operations may be caused by inappropriate relative processing priorities. If adjusting the relative processing priorities of these operations reduces such waiting time, the operations in this critical segment can be processed more compactly. Therefore, the length of the critical path may be shortened, and $C_{\max}(\pi') < C_{\max}(\pi)$ may hold.

Property 6: For a solution π , assume that operation $O_{i,j} \in \Pi$ is processed on workstation w with mode r , where $r < NR_{i,j,w}$. If increasing the processing mode from r to $r' \in NR_{i,j,w}$, $r' > r$, reduces the processing time of $O_{i,j}$ and shortens the critical path, then the obtained solution π' cannot be dominated by π .

Proof: A larger multi-robot collaborative processing mode usually shortens the processing time of an operation. Since $O_{i,j}$ lies on the critical path, reducing its processing time may shorten the length of Π . In this case, $C_{\max}(\pi') < C_{\max}(\pi)$. Although a larger processing mode increase the processing energy consumption rate, the makespan objective is improved. Therefore, π' cannot be dominated by π , because π' is better than π in at least the makespan.

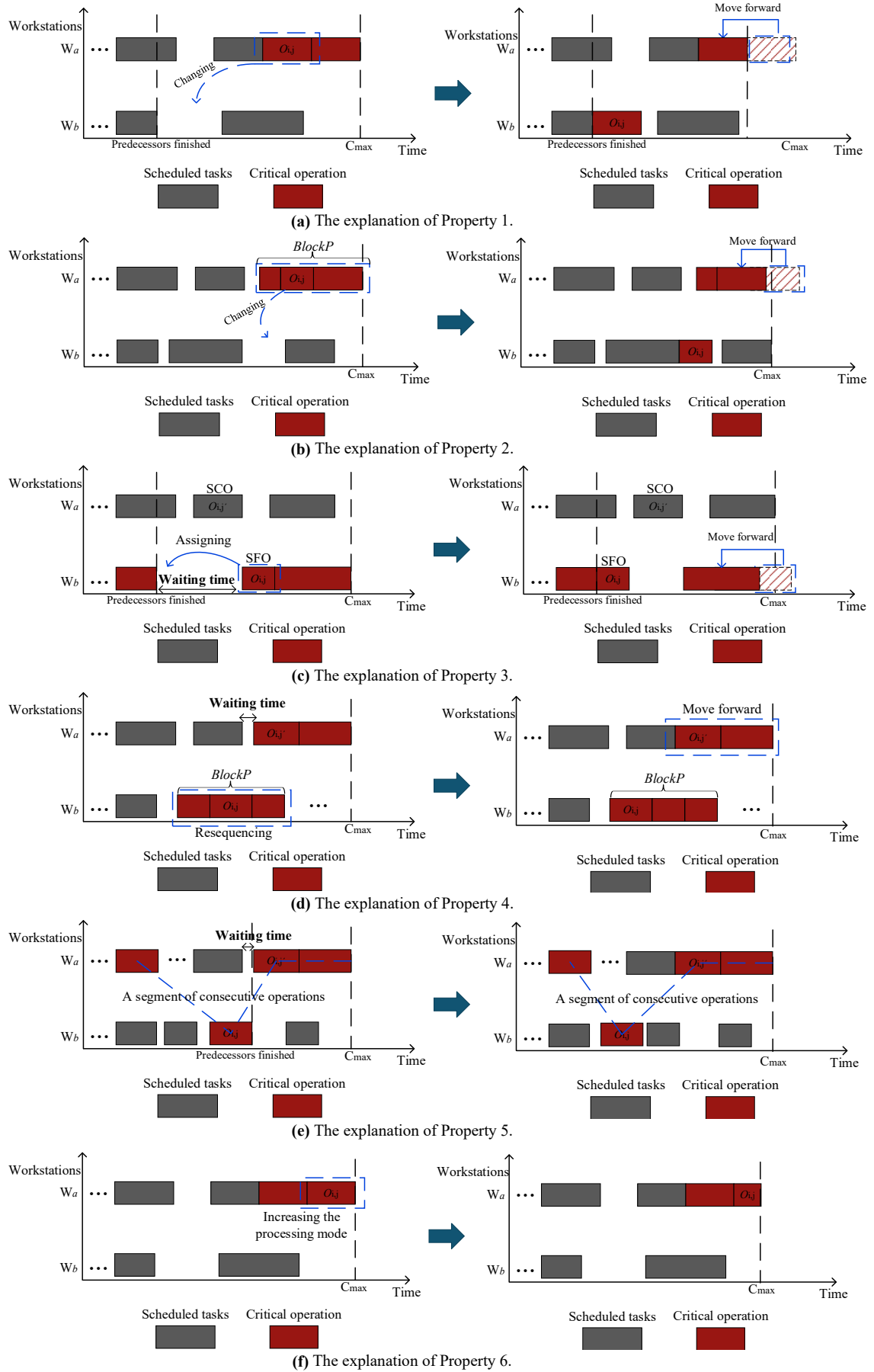


Fig. S-4. Explanation of the six properties.

E. Neighborhood Structures

NS1: Based on Property 1, select the operation on the critical path Π . If there exists a critical operation $O_{i,j} \in \Pi$ such that changing its workstation from w to another eligible workstation $w' \in SEW_{i,j}$ makes its completion time earlier, then replace its current workstation with w' and return the new solution. If more than one critical operation satisfies this condition, select the one with the largest completion time reduction.

NS2: Based on Property 2, select the critical block $BlockP$ on the critical path. If more than one $BlockP$ exists, choose the one with the largest number of operations. Then, search for an operation $O_{i,j} \in BlockP$ such that reassigning it to another eligible workstation can shorten the length of $BlockP$ without delaying the following critical operation. If such an operation exists, change its workstation and return the new solution.

NS3: Based on Property 3, select the SFO on the critical path Π . If there exists a critical SFO $O_{i,j} \in \Pi$ such that assigning it a more suitable relative processing priority makes it start earlier, then adjust its relative processing priority and return the new solution. If more than one critical SFO satisfies this condition, select the one with the largest waiting time before processing.

NS4: Based on Property 4, select the critical block $BlockP$ on the critical path. If more than one $BlockP$ exists, choose the one with the largest number of operations. Then, resequencing the operations inside $BlockP$ without changing their workstation and collaborative processing mode selections.

NS5: Based on Property 5, select a segment of consecutive operations on the critical path Π . If adjusting the relative processing priorities of the operations in this segment can reduce the total waiting time between adjacent critical operations, then update their relative processing priorities and return the new solution. If more than one segment satisfies this condition, select the one with the largest total waiting time.

NS6: Based on Property 6, select the operation on the critical path Π . If there exists a critical operation $O_{i,j} \in \Pi$ processed on workstation w with mode $r < NR_{i,j,w}$, then increase its collaborative processing mode to a larger mode $r' \in NR_{i,j,w}$. If more than one such critical operation exists, select the one with the largest potential processing time reduction.

F. Parameter Sensitivity Analysis

The performance of KD-EMMTMA is directly affected by its parameter settings. Therefore, the Taguchi orthogonal experiment is employed to determine an appropriate parameter combination. KD-EMMTMA contains five key parameters, each with three levels: $TPS = \{100, 150, 200\}$, $APS = \{50, 75, 100\}$, $PC = \{0.7, 0.8, 0.9\}$, $PM = \{0.10, 0.15, 0.20\}$, and $KTP = \{0.1, 0.2, 0.3\}$. Accordingly, the orthogonal array $L_{27}(3^5)$ consisting of 27 parameter combinations, is adopted. Based on the main-effect plots of GD, IGD, and HV shown in Fig. S-5, the parameter combination is determined as $TPS=100$, $APS=50$, $PC=0.7$, $PM=0.1$, and $KTP=0.2$.

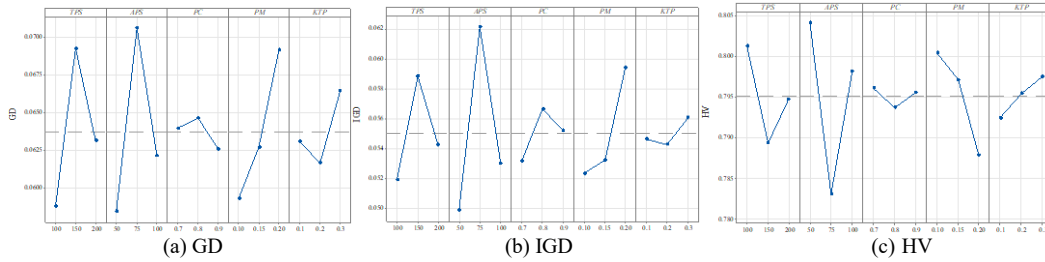


Fig. S-5. Main effects plots of GD, IGD and HV.

G. MILP and CP Model Validation

Since reporting the results for all small-scale instances would require excessive space, only the results for MPSF01 are presented. Table S-II reports the optimal objective values and computational times obtained by the two models for MPSF01. For all 42 ε settings, the MILP and CP models obtain identical $C_{\max}$ and TEC values, thereby generating the same optimal Pareto front. This consistency verifies the correctness of both models for the EE-MRC-FJSPDS.

TABLE S-II
OPTIMAL PARETO SOLUTIONS OF MPSF01

ε	MILP			CP		
	$C_{\max}$	TEC	Time	$C_{\max}$	TEC	Time
87	86	4937	0.05	86	4937	0.10
89	89	4848	0.23	89	4848	0.32
90	90	4745	0.24	90	4745	0.16
95	95	4510	0.27	95	4510	0.25
98	98	4316	0.35	98	4316	0.20
103	103	4221	0.40	103	4221	0.32
109	109	4063	0.31	109	4063	0.90
110	110	4060	1.07	110	4060	0.66
111	111	3933	0.69	111	3933	0.19
113	113	3842	0.36	113	3842	0.22
115	115	3830	0.27	115	3830	0.33
118	118	3747	0.75	118	3747	0.68
119	119	3684	0.56	119	3684	0.62
120	120	3605	0.24	120	3605	0.63
122	122	3514	0.28	122	3514	0.70
127	127	3419	0.21	127	3419	0.36
130	130	3395	0.35	130	3395	0.50
133	133	3302	0.32	133	3302	0.48
137	137	3214	0.11	137	3214	0.33
140	140	3187	0.41	140	3187	0.42
143	143	3115	0.19	143	3115	0.33
148	148	3068	0.33	148	3068	0.66
150	150	2982	0.43	150	2982	0.48
156	156	2922	0.20	156	2922	0.38
157	157	2852	0.13	157	2852	0.28
166	166	2801	0.39	166	2801	0.55
167	167	2782	0.39	167	2782	0.63
170	170	2620	0.16	170	2620	0.16
179	179	2569	0.23	179	2569	0.19
180	180	2472	0.16	180	2472	0.11
191	186	2442	0.24	186	2442	0.23
193	193	2264	0.12	193	2264	0.08
196	196	2243	0.10	196	2243	0.09
199	199	2210	0.36	199	2210	0.07
209	209	2140	0.35	209	2140	0.08
212	212	2131	0.04	212	2131	0.11
213	213	2112	0.09	213	2112	0.13
222	222	2061	0.24	222	2061	0.04
223	223	2042	0.25	223	2042	0.08
225	225	1965	0.24	225	1965	0.05
237	237	1860	0.21	237	1860	0.06
252	252	1839	0.17	252	1839	0.07

H. Ablation Experiments

TABLE S-III
COMPARISON OF GD RESULTS FOR KD-EMMTMA AND ITS FIVE VARIANTS

Instances	KD-EMMTMA-1	KD-EMMTMA-2	KD-EMMTMA-3	KD-EMMTMA-4	KD-EMMTMA-5	KD-EMMTMA
MPSF01	0.004587	0.002711	0.005832	0.00205	0.006861	0.003625
MPSF02	0.01572	0.018221	0.01469	0.015778	0.018424	0.032234
MPSF03	0.01144	0.02045	0.014522	0.015181	0.021568	0.00594
MPSF04	0.022865	0.015096	0.048893	0.027202	0.037584	0.018784
MPSF05	0.023851	0.026963	0.024556	0.019013	0.02439	0.015652
MPSF06	0.022256	0.021379	0.027296	0.012572	0.024421	0.012832
MPSF07	0.022036	0.031564	0.035517	0.023699	0.028683	0.020354
MPSF08	0.02907	0.032219	0.025476	0.044092	0.03435	0.031415
MPSF09	0.055164	0.051281	0.028667	0.029682	0.046667	0.008011
MPSF10	0.018976	0.027273	0.007684	0.014817	0.020049	0.01763
PSF01	0.017014	0.032601	0.036043	0.021594	0.042661	0.015686
PSF02	0.027313	0.061316	0.017126	0.028632	0.032365	0.019928
PSF03	0.026587	0.024765	0.029831	0.022564	0.026457	0.012282
PSF04	0.031333	0.043875	0.029555	0.033315	0.049772	0.008916
PSF05	0.036635	0.029017	0.031464	0.038267	0.036801	0.020416
PSF06	0.040106	0.052698	0.01585	0.042651	0.044615	0.004007
PSF07	0.020876	0.01696	0.028782	0.032083	0.03142	0.012481
PSF08	0.010863	0.020772	0.026942	0.048675	0.035226	0.020977
PSF09	0.063738	0.048898	0.037893	0.065612	0.004374	0.032712
PSF10	0.030686	0.018775	0.030451	0.035943	0.03043	0.00597
PSF11	0.068655	0.085249	0.06057	0.045206	0.039555	0.061101
PSF12	0.028162	0.073165	0.064495	0.070836	0.051098	0.023647
PSF13	0.020194	0.013277	0.030092	0.03123	0.070963	0.032673
PSF14	0.058459	0.033047	0.033701	0.043677	0.038214	0.012766
PSF15	0.047537	0.050837	0.01562	0.023238	0.037675	0.022732
PSF16	0.020617	0.035938	0.026678	0.032039	0.038994	0.031393
PSF17	0.040411	0.01276	0.110283	0.063629	0.040549	0.034919
PSF18	0.031637	0.028194	0.040029	0.022987	0.051292	0.023178
PSF19	0.032153	0.014978	0.050684	0.040651	0.046281	0.042999
PSF20	0.029174	0.022466	0.030918	0.030042	0.030878	0.030115
PSF21	0.029996	0.026201	0.037866	0.035893	0.035038	0.011647
PSF22	0.021404	0.031493	0.016982	0.036882	0.041704	0.009739
PSF23	0.01543	0.059958	0.100117	0.070689	0.085859	0.05608
PSF24	0.021574	0.041056	0.05451	0.065097	0.069802	0.022961
PSF25	0.017916	0.051865	0.021093	0.020609	0.035648	0.046724
PSF26	0.052015	0.050066	0.035956	0.035138	0.082261	0.011261
PSF27	0.029223	0.041966	0.039788	0.043133	0.062357	0.036053
PSF28	0.024494	0.046856	0.037032	0.019244	0.030934	0.024324
PSF29	0.028873	0.037678	0.058076	0.073581	0.076707	0.005867
PSF30	0.029039	0.063751	0.032665	0.05623	0.057825	0.020063
PSF31	0.007625	0.022914	0.047483	0.050403	0.024439	0.012105
PSF32	0.032664	0.074977	0.033731	0.064306	0.071297	0.004644
PSF33	0.075618	0.077809	0.082657	0.139733	0.094802	0.000564
PSF34	0.033728	0.008064	0.030579	0.102238	0.068668	0.040076
PSF35	0.052811	0.038789	0.026016	0.018317	0.065996	0.029525
PSF36	0.034802	0.032344	0.057714	0.053897	0.096447	0.026782
PSF37	0.008005	0.064753	0.041864	0.033858	0.057783	0.032798
PSF38	0.027524	0.011148	0.028198	0.039504	0.061602	0.012446
PSF39	0.036976	0.062585	0.056376	0.052957	0.058236	0.01723
PSF40	0.046983	0.004212	0.043988	0.028286	0.044678	0.026643
PSF41	0.022164	0.028554	0.019336	0.028923	0.047001	0.05249
PSF42	0.032955	0.022864	0.057212	0.034426	0.057808	0.005184
PSF43	0.034479	0.03269	0.026493	0.049217	0.03916	0.010326
PSF44	0.015653	0.064296	0.071441	0.059669	0.039015	0.050637
PSF45	0.030095	0.041373	0.058003	0.038433	0.070311	0.016874
PSF46	0.019203	0.054147	0.013463	0.040456	0.045876	0.027927
PSF47	0.024474	0.027106	0.03218	0.038482	0.029634	0.021081
PSF48	0.027034	0.042853	0.045613	0.030327	0.056351	0.01894
PSF49	0.011401	0.039264	0.031932	0.02158	0.053717	0.027899
PSF50	0.010858	0.046311	0.042487	0.047115	0.042812	0.023329
PSF51	0.001228	0.048586	0.072423	0.043907	0.075135	0.033399
PSF52	0.02658	0.034764	0.0563	0.041938	0.048042	0.022373
PSF53	0.015922	0.05143	0.015527	0.052559	0.015673	0.026596
PSF54	0.025767	0.042264	0.051251	0.061287	0.052757	0.006432
PSF55	0.038004	0.043849	0.085144	0.094915	0.075102	0.006881
PSF56	0.024065	0.034796	0.083407	0.073935	0.015212	0.030504
PSF57	0.020831	0.042655	0.01868	0.05842	0.057302	0.011611
PSF58	0.045611	0.036665	0.067444	0.068093	0.071594	0.00807
PSF59	0.026263	0.039161	0.044193	0.069111	0.061644	0.007805

PSF60	0.011556	0.044396	0.046111	0.062156	0.058182	0.045885
PSF61	0.020416	0.065162	0.077419	0.072457	0.079646	0.035002
PSF62	0.005726	0.026451	0.049597	0.090113	0.055017	0.025857
PSF63	0.007782	0.021127	0.057449	0.085206	0.081819	0.026015
PSF64	0.013636	0.021717	0.037069	0.116276	0.070826	0.044413
PSF65	0.031125	0.04375	0.068209	0.11903	0.020139	0.032958
PSF66	0.017801	0.026534	0.035197	0.089673	0.06037	0.035452
PSF67	0.019565	0.046652	0.066487	0.075782	0.043437	0.021501
PSF68	0.024853	0.056771	0.034044	0.075686	0.067547	0.02685
PSF69	0.042251	0.059935	0.059348	0.088972	0.090193	0.01949
PSF70	0.007806	0.07522	0.084619	0.05802	0.059462	0.021405
PSF71	0.011977	0.042287	0.036796	0.069535	0.059063	0.040968
PSF72	0.032261	0.057907	0.035045	0.081658	0.058527	0.009029
PSF73	0.022665	0.038852	0.040179	0.092902	0.069788	0.025842
PSF74	0.013628	0.029454	0.042956	0.046101	0.033311	0.026536
PSF75	0.018553	0.025781	0.021656	0.074742	0.056892	0.023271
PSF76	0.010973	0.061136	0.040836	0.061388	0.087249	0.046749
PSF77	0.044973	0.021035	0.0817	0.092435	0.099919	0.040861
PSF78	0.049772	0.049131	0.087071	0.09657	0.071769	0.034514
PSF79	0.026807	0.056445	0.050775	0.035949	0.055112	0.018421
PSF80	0.014338	0.035215	0.050163	0.053923	0.049692	0.00601
PSF81	0.026625	0.025951	0.066211	0.103036	0.062568	0.023749
PSF82	0.013444	0.062651	0.027403	0.070458	0.074719	0.034889
PSF83	0.011154	0.01926	0.037433	0.067992	0.057078	0.013298
PSF84	0.020327	0.06007	0.044114	0.073062	0.040051	0.031285
PSF85	0.029652	0.047905	0.072879	0.110737	0.011256	0.035083
PSF86	0.021774	0.022723	0.035432	0.065744	0.070266	0.054239
PSF87	0.040951	0.045925	0.065401	0.063473	0.035918	0.037124
PSF88	0.053991	0.057055	0.069881	0.093714	0.052198	0.021824
PSF89	0.030369	0.033417	0.055878	0.06602	0.0849	0.008617
PSF90	0.023558	0.021682	0.070635	0.053577	0.058418	0.031574
PSF91	0.032455	0.046022	0.049538	0.066559	0.047606	0.016628
PSF92	0.053608	0.0432	0.04741	0.056329	0.061566	0.018185
PSF93	0.04784	0.029924	0.030432	0.035212	0.050443	0.02954
PSF94	0.02475	0.070902	0.046723	0.080088	0.072171	0.0284
PSF95	0.02066	0.023034	0.045446	0.059281	0.04571	0.038033
PSF96	0.034438	0.066793	0.062688	0.081626	0.039876	0.034605
PSF97	0.042734	0.078406	0.071618	0.081513	0.085283	0.02211
PSF98	0.016384	0.073025	0.055758	0.054445	0.069873	0.049562
PSF99	0.047356	0.054009	0.064086	0.091825	0.071805	0.012005
PSF100	0.022315	0.048396	0.050122	0.070305	0.067607	0.00597
Mean	0.02773	0.04022	0.044792	0.05503	0.051874	0.024046

TABLE S-IV
COMPARISON OF IGD RESULTS FOR KD-EMMTMA AND ITS FIVE VARIANTS

Instances	KD- EMMTMA-1	KD- EMMTMA-2	KD- EMMTMA-3	KD- EMMTMA-4	KD- EMMTMA-5	KD- EMMTMA
MPSF01	0.011634	0.011332	0.010726	0.008552	0.005814	0.006357
MPSF02	0.032934	0.037749	0.034927	0.015967	0.030319	0.023141
MPSF03	0.025195	0.02453	0.022369	0.014621	0.031704	0.030115
MPSF04	0.025014	0.026314	0.0428	0.023117	0.050042	0.024181
MPSF05	0.033396	0.040813	0.028618	0.018471	0.04359	0.025389
MPSF06	0.028605	0.032826	0.034173	0.014496	0.04446	0.032327
MPSF07	0.024662	0.029716	0.030067	0.023874	0.037477	0.021703
MPSF08	0.043559	0.043321	0.028156	0.038435	0.040355	0.029406
MPSF09	0.033505	0.073062	0.062341	0.023586	0.090155	0.058632
MPSF10	0.024792	0.040735	0.014282	0.013787	0.028029	0.021262
PSF01	0.048743	0.076199	0.061543	0.020215	0.06592	0.031426
PSF02	0.054687	0.067025	0.035414	0.025598	0.066678	0.032734
PSF03	0.053701	0.059312	0.047126	0.020994	0.052481	0.040742
PSF04	0.045605	0.081767	0.056127	0.031847	0.085876	0.03421
PSF05	0.028025	0.061536	0.030722	0.038279	0.037568	0.03844
PSF06	0.062529	0.063632	0.044089	0.039072	0.054298	0.053137
PSF07	0.032296	0.073181	0.043567	0.030057	0.037883	0.021259
PSF08	0.032102	0.096585	0.035736	0.042727	0.045412	0.02613
PSF09	0.025235	0.090066	0.036166	0.049632	0.012458	0.050936
PSF10	0.015827	0.055425	0.036526	0.031176	0.029507	0.029714
PSF11	0.040797	0.066367	0.050785	0.042733	0.041077	0.053669
PSF12	0.015558	0.092399	0.067779	0.077226	0.054057	0.040734
PSF13	0.043398	0.055919	0.030135	0.038923	0.075163	0.02913
PSF14	0.019873	0.081554	0.032286	0.046512	0.043953	0.056003
PSF15	0.054523	0.073927	0.023441	0.022879	0.070446	0.09823
PSF16	0.036979	0.122171	0.038886	0.026159	0.045791	0.023165
PSF17	0.055831	0.086043	0.110222	0.057031	0.038171	0.045679
PSF18	0.033843	0.07109	0.041572	0.020108	0.04894	0.038556
PSF19	0.045401	0.043505	0.05692	0.037266	0.057999	0.029229
PSF20	0.042672	0.098426	0.041229	0.029772	0.038451	0.028409
PSF21	0.033234	0.102738	0.066946	0.034049	0.044887	0.051287
PSF22	0.02125	0.068703	0.035888	0.032464	0.048243	0.024645
PSF23	0.049703	0.092315	0.087259	0.051677	0.064611	0.024229
PSF24	0.049726	0.095272	0.056802	0.051597	0.075867	0.033159
PSF25	0.056617	0.105787	0.061119	0.023849	0.058486	0.042902
PSF26	0.065353	0.102765	0.073419	0.021874	0.059299	0.058582
PSF27	0.039418	0.077546	0.049784	0.038348	0.066173	0.036452
PSF28	0.0469	0.114232	0.045306	0.016509	0.051751	0.029015
PSF29	0.053916	0.137494	0.054252	0.068196	0.082151	0.03161
PSF30	0.030808	0.11562	0.030556	0.040394	0.061292	0.030567
PSF31	0.041716	0.105002	0.063516	0.049354	0.054186	0.01685
PSF32	0.01352	0.116125	0.040237	0.060815	0.061452	0.03465
PSF33	0.001679	0.171716	0.069979	0.125104	0.071702	0.064745
PSF34	0.043742	0.110987	0.033279	0.095685	0.060882	0.036649
PSF35	0.027871	0.093215	0.033581	0.024359	0.068975	0.059099
PSF36	0.035926	0.076659	0.05152	0.045604	0.090819	0.038228
PSF37	0.071156	0.082622	0.075891	0.024759	0.072243	0.068284
PSF38	0.058336	0.084794	0.060643	0.025199	0.080274	0.062552
PSF39	0.043116	0.133513	0.066954	0.041508	0.079131	0.05614
PSF40	0.047016	0.112421	0.10642	0.022566	0.063669	0.055576
PSF41	0.077973	0.146256	0.056524	0.029556	0.082744	0.044382
PSF42	0.016002	0.099439	0.05834	0.033715	0.06655	0.043141
PSF43	0.031153	0.062053	0.029432	0.039712	0.038958	0.035842
PSF44	0.10084	0.092977	0.105961	0.04505	0.081838	0.027729
PSF45	0.038649	0.098453	0.060864	0.027248	0.051347	0.03384
PSF46	0.101805	0.217479	0.079745	0.030794	0.098674	0.06213
PSF47	0.074295	0.109682	0.042817	0.035911	0.040314	0.025572
PSF48	0.041021	0.102452	0.072641	0.026036	0.064251	0.047717
PSF49	0.049269	0.088242	0.05124	0.015953	0.055876	0.05051
PSF50	0.02278	0.101414	0.047442	0.044306	0.050906	0.028497
PSF51	0.044919	0.089766	0.076872	0.03897	0.074341	0.021658
PSF52	0.046798	0.098098	0.063969	0.033875	0.07903	0.04084
PSF53	0.046785	0.108428	0.040451	0.051878	0.037274	0.031427
PSF54	0.034182	0.145141	0.051002	0.065015	0.071009	0.03098
PSF55	0.036227	0.11868	0.079641	0.089093	0.074733	0.042605
PSF56	0.031037	0.155535	0.064365	0.065161	0.030742	0.027215
PSF57	0.022003	0.156648	0.050529	0.05301	0.063997	0.02225
PSF58	0.012737	0.136267	0.060643	0.057976	0.071781	0.041121
PSF59	0.021924	0.146174	0.040474	0.068596	0.068397	0.035111
PSF60	0.041038	0.157199	0.056066	0.053527	0.056675	0.020646
PSF61	0.029753	0.179963	0.067019	0.057317	0.074965	0.019784

PSF62	0.031495	0.119657	0.050753	0.08025	0.04817	0.016641
PSF63	0.030444	0.228835	0.055281	0.081033	0.071084	0.017307
PSF64	0.038533	0.197553	0.042015	0.116996	0.062132	0.024152
PSF65	0.035649	0.208215	0.064468	0.111789	0.03148	0.034639
PSF66	0.034847	0.19786	0.030923	0.082204	0.057713	0.019629
PSF67	0.024819	0.168691	0.061157	0.066059	0.038895	0.020441
PSF68	0.024823	0.254435	0.037922	0.074444	0.059728	0.027259
PSF69	0.021939	0.148637	0.056107	0.069252	0.080484	0.027074
PSF70	0.020631	0.194454	0.069144	0.042146	0.050735	0.016437
PSF71	0.043088	0.193182	0.034727	0.067138	0.052887	0.016261
PSF72	0.010006	0.189987	0.028037	0.067488	0.059605	0.041398
PSF73	0.025722	0.264488	0.039849	0.088483	0.067821	0.030328
PSF74	0.032961	0.226198	0.045414	0.04237	0.032479	0.019707
PSF75	0.026308	0.230437	0.021498	0.059332	0.046344	0.025896
PSF76	0.053776	0.187741	0.056152	0.050837	0.092317	0.013367
PSF77	0.033886	0.164842	0.078467	0.092501	0.098187	0.0406
PSF78	0.056343	0.101436	0.076849	0.087353	0.055546	0.061785
PSF79	0.019505	0.172776	0.040328	0.030766	0.047572	0.023628
PSF80	0.024678	0.201043	0.039848	0.038557	0.052464	0.022525
PSF81	0.028097	0.16803	0.055461	0.086848	0.056362	0.019467
PSF82	0.036804	0.2823	0.023362	0.068402	0.080527	0.015401
PSF83	0.026009	0.201626	0.046886	0.056689	0.061025	0.014069
PSF84	0.034487	0.186424	0.044699	0.068862	0.040958	0.025662
PSF85	0.031133	0.129617	0.056197	0.09447	0.031483	0.022063
PSF86	0.043942	0.182161	0.041458	0.069025	0.067333	0.020828
PSF87	0.035489	0.243117	0.05774	0.046039	0.031775	0.035481
PSF88	0.024671	0.123096	0.050173	0.073394	0.038228	0.040518
PSF89	0.017059	0.160435	0.046556	0.055314	0.073374	0.02922
PSF90	0.033609	0.139609	0.065884	0.040356	0.051932	0.026124
PSF91	0.039465	0.143535	0.051568	0.068211	0.053955	0.033051
PSF92	0.023737	0.202493	0.04606	0.049815	0.059573	0.041404
PSF93	0.026087	0.219783	0.04354	0.03366	0.042624	0.045941
PSF94	0.030587	0.226198	0.046988	0.073251	0.068632	0.02576
PSF95	0.042835	0.24312	0.040295	0.054382	0.047981	0.021547
PSF96	0.028484	0.148235	0.050959	0.061987	0.033884	0.022635
PSF97	0.025008	0.182985	0.060228	0.07172	0.077759	0.033466
PSF98	0.049598	0.162867	0.052825	0.041272	0.066225	0.020033
PSF99	0.029862	0.184423	0.058586	0.062591	0.06162	0.034852
PSF100	0.020923	0.115984	0.045197	0.042518	0.063736	0.023775
Mean	0.036731	0.126208	0.050561	0.048614	0.056993	0.033751

TABLE S-V
COMPARISON OF HV RESULTS FOR KD-EMMTMA AND ITS FIVE VARIANTS

Instances	KD-EMMTMA-1	KD-EMMTMA-2	KD-EMMTMA-3	KD-EMMTMA-4	KD-EMMTMA-5	KD-EMMTMA
MPSF01	0.909205	0.911338	0.911571	0.914	0.911127	0.912339
MPSF02	0.77688	0.789473	0.801542	0.812212	0.792143	0.813517
MPSF03	0.853767	0.847127	0.853476	0.870231	0.831922	0.844566
MPSF04	0.829001	0.819469	0.792001	0.830069	0.782571	0.820807
MPSF05	0.833274	0.82138	0.820475	0.84492	0.814586	0.829952
MPSF06	0.830368	0.822409	0.81253	0.852293	0.813329	0.827247
MPSF07	0.892548	0.869017	0.877014	0.895593	0.865316	0.895752
MPSF08	0.888676	0.872358	0.894274	0.885553	0.877734	0.883517
MPSF09	0.851124	0.773621	0.820651	0.840925	0.775076	0.79814
MPSF10	0.735577	0.698591	0.760604	0.75652	0.728544	0.739494
PSF01	0.735572	0.682821	0.710386	0.753085	0.700755	0.753596
PSF02	0.85339	0.78463	0.879595	0.887055	0.849534	0.879015
PSF03	0.757193	0.726724	0.75723	0.791764	0.774703	0.760695
PSF04	0.81683	0.730939	0.776863	0.809977	0.738652	0.807916
PSF05	0.864947	0.798958	0.852442	0.83909	0.841811	0.83588
PSF06	0.784695	0.725085	0.803153	0.778549	0.762932	0.769131
PSF07	0.807664	0.746856	0.790625	0.796553	0.785425	0.815149
PSF08	0.84444	0.771337	0.847268	0.818474	0.81983	0.86205
PSF09	0.783666	0.698739	0.767376	0.73909	0.821054	0.741303
PSF10	0.813441	0.74877	0.786482	0.79527	0.80027	0.798404
PSF11	0.933307	0.849228	0.935726	0.966984	0.972601	0.949327
PSF12	0.855298	0.739212	0.799474	0.778298	0.812558	0.833832
PSF13	0.782167	0.745554	0.809272	0.786325	0.73643	0.800888
PSF14	0.823	0.728509	0.808029	0.802241	0.789946	0.76956
PSF15	0.88955	0.868141	0.917401	0.922622	0.884253	0.863479
PSF16	0.837698	0.712804	0.83392	0.851154	0.830319	0.873593
PSF17	0.79583	0.725153	0.702272	0.77716	0.812647	0.816771
PSF18	0.807239	0.743151	0.7923	0.832346	0.78348	0.798938
PSF19	0.767447	0.74298	0.753397	0.778564	0.758788	0.809065
PSF20	0.803754	0.715292	0.807873	0.803309	0.809937	0.815005
PSF21	0.794746	0.681459	0.72815	0.773009	0.759319	0.767875
PSF22	0.818867	0.719825	0.789172	0.786205	0.764654	0.805533
PSF23	0.790931	0.679604	0.739573	0.79041	0.748827	0.839834
PSF24	0.848446	0.705534	0.804603	0.802496	0.765699	0.849864
PSF25	0.768177	0.673506	0.779142	0.824214	0.777583	0.801343
PSF26	0.800328	0.66305	0.75718	0.788946	0.754699	0.764606
PSF27	0.811037	0.713356	0.801384	0.806845	0.765093	0.83431
PSF28	0.755572	0.600828	0.75611	0.789388	0.760546	0.777708
PSF29	0.811527	0.631016	0.77872	0.751195	0.732088	0.816769
PSF30	0.761742	0.566344	0.750554	0.721026	0.70582	0.763154
PSF31	0.798183	0.651892	0.738779	0.762497	0.773814	0.821315
PSF32	0.834965	0.656813	0.800471	0.774585	0.774524	0.822648
PSF33	0.907337	0.67139	0.800997	0.733419	0.795711	0.816048
PSF34	0.849614	0.716356	0.859812	0.760816	0.82135	0.853218
PSF35	0.823063	0.723775	0.823942	0.834628	0.768147	0.78283
PSF36	0.848388	0.685964	0.779506	0.806083	0.706204	0.808044
PSF37	0.821136	0.683213	0.77335	0.825823	0.766397	0.833754
PSF38	0.824406	0.767947	0.796464	0.841896	0.75163	0.807048
PSF39	0.852446	0.654413	0.789058	0.820425	0.784181	0.812026
PSF40	0.804813	0.726257	0.716308	0.813251	0.761363	0.771373
PSF41	0.741858	0.658706	0.778608	0.797577	0.745702	0.797468
PSF42	0.849964	0.711643	0.763342	0.818919	0.766388	0.815571
PSF43	0.821838	0.714531	0.813562	0.771104	0.791034	0.793316
PSF44	0.783864	0.697241	0.741096	0.792756	0.822219	0.847873
PSF45	0.866441	0.718803	0.807525	0.849728	0.822302	0.855669
PSF46	0.788616	0.64052	0.808769	0.820924	0.776152	0.827137
PSF47	0.791472	0.692793	0.78107	0.795119	0.788822	0.8155
PSF48	0.783727	0.649052	0.730175	0.780824	0.736337	0.773346
PSF49	0.796433	0.686895	0.779855	0.833557	0.773664	0.811173
PSF50	0.861043	0.724721	0.81969	0.827506	0.794038	0.862112
PSF51	0.843097	0.69157	0.76553	0.823863	0.767215	0.872878
PSF52	0.80906	0.711235	0.780629	0.810114	0.770869	0.828796
PSF53	0.872833	0.74035	0.870447	0.854517	0.857907	0.896425
PSF54	0.761448	0.58984	0.73562	0.725308	0.70085	0.773479
PSF55	0.866922	0.706987	0.799397	0.793774	0.818774	0.865402
PSF56	0.791374	0.653964	0.728716	0.733988	0.792932	0.805361
PSF57	0.830262	0.629896	0.804353	0.765715	0.758293	0.823605
PSF58	0.856403	0.659242	0.772183	0.778894	0.746933	0.80937
PSF59	0.85473	0.698509	0.807261	0.772916	0.768896	0.843468
PSF60	0.804304	0.639042	0.789935	0.775411	0.770489	0.851002
PSF61	0.85068	0.559215	0.770546	0.795076	0.766552	0.8668

PSF62	0.841327	0.601049	0.803133	0.752801	0.805678	0.879738
PSF63	0.856653	0.592602	0.8189	0.792509	0.804586	0.899613
PSF64	0.81731	0.624102	0.798746	0.687196	0.772377	0.847922
PSF65	0.809814	0.613586	0.763711	0.700708	0.841043	0.835727
PSF66	0.839575	0.664863	0.853567	0.769288	0.806573	0.870694
PSF67	0.692308	0.490886	0.647034	0.629196	0.67854	0.716438
PSF68	0.771006	0.458608	0.740813	0.691223	0.709	0.767162
PSF69	0.7805	0.484307	0.727852	0.687535	0.684017	0.761999
PSF70	0.772243	0.490365	0.681379	0.717521	0.70652	0.785647
PSF71	0.717951	0.457549	0.736552	0.674256	0.695163	0.778707
PSF72	0.839745	0.554403	0.817481	0.752703	0.763314	0.805696
PSF73	0.768893	0.457936	0.740439	0.661119	0.695902	0.754682
PSF74	0.7335	0.483202	0.715756	0.707865	0.736789	0.752263
PSF75	0.769941	0.521194	0.77091	0.710285	0.737936	0.768454
PSF76	0.76135	0.520665	0.757581	0.743336	0.687013	0.814022
PSF77	0.860103	0.638106	0.782104	0.765032	0.745629	0.844692
PSF78	0.817954	0.623828	0.772434	0.756367	0.793272	0.828246
PSF79	0.838733	0.636935	0.814431	0.831908	0.788454	0.858662
PSF80	0.880541	0.658946	0.851698	0.847019	0.82622	0.891132
PSF81	0.781928	0.519896	0.739689	0.682115	0.730837	0.787829
PSF82	0.813771	0.519646	0.828217	0.758124	0.734729	0.847613
PSF83	0.792331	0.492318	0.745384	0.718499	0.715681	0.801975
PSF84	0.765074	0.503929	0.75205	0.716672	0.769997	0.798298
PSF85	0.767582	0.489438	0.722384	0.664978	0.785881	0.778374
PSF86	0.794346	0.588832	0.803868	0.757853	0.764698	0.835606
PSF87	0.770536	0.55524	0.742513	0.753554	0.787445	0.782266
PSF88	0.829372	0.594876	0.7816	0.740857	0.791769	0.802271
PSF89	0.827947	0.576777	0.78056	0.762193	0.7402	0.812371
PSF90	0.809641	0.595919	0.747179	0.7796	0.777396	0.817767
PSF91	0.736498	0.526057	0.688848	0.654196	0.680082	0.720978
PSF92	0.793184	0.534622	0.747841	0.731086	0.727657	0.746759
PSF93	0.789702	0.576622	0.790841	0.787916	0.775189	0.777709
PSF94	0.790028	0.548965	0.777146	0.732958	0.756311	0.825998
PSF95	0.732834	0.505138	0.748508	0.711883	0.719722	0.771369
PSF96	0.804306	0.546261	0.765534	0.744491	0.800805	0.813735
PSF97	0.7922	0.504743	0.736658	0.713178	0.721146	0.792537
PSF98	0.75932	0.553217	0.764289	0.764694	0.744643	0.820931
PSF99	0.868497	0.566355	0.797607	0.770032	0.79758	0.827321
PSF100	0.854936	0.572753	0.801499	0.797686	0.781535	0.847154
Mean	0.812301	0.658197	0.784941	0.781413	0.774851	0.81613

I. Parameter settings for comparison algorithms

Table S-VI provides the parameter settings of the comparison algorithms, which are tuned using the Taguchi orthogonal experiment.

TABLE S-VI
THE PARAMETERS SETTINGS OF COMPARISON ALGORITHMS

Algorithm	Parameter	Value
IDPSO	Elite particle ratio	0.6
	Chasing particle ratio	0.9
	Population initialization selection probability	0.7
TSCOA	Population size	100
	Crossover rate	0.7
	Mutation rate	0.1
	Iteration allocation parameter	0.1
KEMMA	Population size	200
	Task allocation ratio	0.7
	Random mating probability	0.3
MOMA	Population size	100
	Mutation probability	0.8
	Discount factor	0.9
	Learning rate	0.2
	Greedy factor	0.9
HEA	Knowledge transfer rate	0.1
	Population size	150
	Crossover probability	0.7
	Mutation probability	0.2
	Local search probability	0.45

J. Comparative Experiments

TABLE S-VII
COMPARISON OF GD RESULTS FOR KD-EMMTMA AND EXISTING STATE-OF-THE-ART ALGORITHMS

Instances	IDPSO	TSCOA	KEMMA	MOMA	IMA	KD-EMMTMA
MPSF01	0.014039	0.008307	0.014918	0.014464	0.022042	0.002424
MPSF02	0.017888	0.014711	0.049609	0.040689	0.067276	0.017012
MPSF03	0.04383	0.043439	0.052623	0.039295	0.065917	9.60E-05
MPSF04	0.091388	0.048137	0.092153	0.093466	0.081327	6.15E-05
MPSF05	0.062256	0.063021	0.049681	0.072114	0.137682	0
MPSF06	0.033911	0.059086	0.058285	0.052221	0.076896	0.000996
MPSF07	0.060217	0.036262	0.052327	0.04435	0.118333	0.012244
MPSF08	0.075836	0.061044	0.094938	0.130518	0.132405	0
MPSF09	0.106909	0.118219	0.115829	0.15429	0.155773	0
MPSF10	0.035233	0.029495	0.040833	0.041939	0.045825	0.003141
PSF01	0.085578	0.080299	0.109667	0.101452	0.130429	0
PSF02	0.088148	0.060329	0.080309	0.084474	0.110191	0.00528
PSF03	0.1145	0.114955	0.116288	0.107882	0.125973	0.000315
PSF04	0.087173	0.081574	0.100127	0.107319	0.138441	0
PSF05	0.11405	0.089686	0.086357	0.083553	0.146983	0
PSF06	0.111008	0.109245	0.122476	0.128323	0.126365	0
PSF07	0.098502	0.103395	0.116945	0.090077	0.111995	0
PSF08	0.06012	0.069425	0.095087	0.084717	0.088992	0
PSF09	0.071792	0.083297	0.073965	0.08909	0.114922	0
PSF10	0.046564	0.106962	0.111598	0.119884	0.130657	0
PSF11	0.102928	0.070301	0.126613	0.115973	0.150892	0.005004
PSF12	0.124927	0.142571	0.1113	0.171036	0.174857	0
PSF13	0.093771	0.066847	0.085206	0.112836	0.122517	0
PSF14	0.098157	0.081104	0.105337	0.1392	0.160716	0.000805
PSF15	0.062795	0.056797	0.066319	0.068771	0.0986	0
PSF16	0.106446	0.081199	0.111231	0.111139	0.114042	0
PSF17	0.104174	0.147993	0.160103	0.184972	0.19258	0.002058
PSF18	0.096259	0.085172	0.100613	0.112355	0.126264	0
PSF19	0.102844	0.068977	0.087179	0.098109	0.138174	0.00333
PSF20	0.049677	0.089703	0.074074	0.076766	0.081444	0
PSF21	0.111891	0.128656	0.111009	0.137198	0.134445	0
PSF22	0.114058	0.087679	0.105981	0.123824	0.116554	0
PSF23	0.126001	0.103021	0.096923	0.102617	0.122536	0.001092
PSF24	0.118996	0.148573	0.11559	0.149429	0.157486	0
PSF25	0.092976	0.072047	0.077119	0.111187	0.110376	0.001556
PSF26	0.136282	0.152639	0.167976	0.136081	0.159431	0.002413
PSF27	0.109259	0.121316	0.093283	0.148347	0.11558	0.000423
PSF28	0.118766	0.086573	0.110763	0.122699	0.115062	0.000911
PSF29	0.11457	0.132645	0.086966	0.13473	0.176683	0
PSF30	0.121613	0.126543	0.13916	0.140143	0.122578	0.004988
PSF31	0.062487	0.113239	0.122227	0.14537	0.12688	0
PSF32	0.125715	0.085643	0.115979	0.108674	0.169267	0
PSF33	0.128101	0.153375	0.186299	0.216736	0.222998	0
PSF34	0.11382	0.111271	0.113007	0.120304	0.125257	0.003367
PSF35	0.105848	0.116283	0.07737	0.119943	0.134777	0
PSF36	0.112588	0.161801	0.103553	0.144987	0.115893	0
PSF37	0.124242	0.09919	0.083806	0.153906	0.154199	0
PSF38	0.097985	0.142085	0.123445	0.126553	0.191868	1.59E-05
PSF39	0.115255	0.114039	0.105459	0.151153	0.18815	0
PSF40	0.099025	0.119375	0.101042	0.107653	0.168122	0
PSF41	0.076279	0.100628	0.091306	0.078921	0.110273	0
PSF42	0.121873	0.14443	0.100903	0.143412	0.183878	0
PSF43	0.112798	0.115006	0.147998	0.129397	0.154019	0
PSF44	0.117473	0.111284	0.120923	0.104483	0.120336	0
PSF45	0.114976	0.113539	0.130463	0.178522	0.16455	0
PSF46	0.10666	0.118374	0.143097	0.11695	0.186369	0
PSF47	0.101349	0.10946	0.117843	0.113134	0.152996	0
PSF48	0.10284	0.139962	0.097699	0.155022	0.159503	0
PSF49	0.12246	0.110065	0.096851	0.118918	0.13153	0.00022
PSF50	0.094648	0.126358	0.113608	0.11276	0.132151	0
PSF51	0.159539	0.157025	0.170867	0.154642	0.14753	0
PSF52	0.122975	0.121143	0.121178	0.130999	0.111638	0
PSF53	0.132256	0.116314	0.120578	0.156511	0.096645	0
PSF54	0.093738	0.117274	0.138792	0.123049	0.130433	0
PSF55	0.165928	0.164404	0.134866	0.176674	0.184972	0
PSF56	0.117155	0.080623	0.116343	0.121519	0.123521	0.012097
PSF57	0.078407	0.103114	0.121758	0.121709	0.130413	0
PSF58	0.123879	0.126895	0.117802	0.090226	0.146434	0.000825
PSF59	0.109329	0.123927	0.107034	0.113033	0.149831	0
PSF60	0.070075	0.101587	0.109665	0.086548	0.128885	0.00483

PSF61	0.111486	0.080149	0.12307	0.121793	0.14937	0
PSF62	0.165535	0.100472	0.129118	0.164002	0.156763	0
PSF63	0.138634	0.105554	0.10255	0.124506	0.161014	0
PSF64	0.095473	0.08903	0.095187	0.119432	0.139921	0.001052
PSF65	0.154963	0.137036	0.119045	0.161416	0.13968	0
PSF66	0.152428	0.120976	0.06678	0.112454	0.198759	0
PSF67	0.092482	0.0892	0.108841	0.111319	0.138244	0.001617
PSF68	0.091411	0.089045	0.083923	0.112816	0.101805	0.003336
PSF69	0.121599	0.117919	0.117319	0.107981	0.087307	0
PSF70	0.085467	0.100778	0.087954	0.117165	0.090864	0
PSF71	0.081501	0.098486	0.081942	0.09008	0.118015	0.005234
PSF72	0.127281	0.153764	0.09446	0.109836	0.148328	0
PSF73	0.119131	0.111712	0.097586	0.114333	0.138264	0
PSF74	0.077595	0.077932	0.085322	0.088311	0.081549	0
PSF75	0.082029	0.05768	0.080642	0.098927	0.107166	0.002151
PSF76	0.125436	0.138908	0.096241	0.134237	0.111031	0
PSF77	0.06882	0.170787	0.126022	0.123469	0.21733	0.006299
PSF78	0.166663	0.101388	0.14047	0.132265	0.190763	0
PSF79	0.100522	0.072707	0.14176	0.098304	0.129574	0
PSF80	0.127022	0.105789	0.150317	0.129968	0.183514	0
PSF81	0.14269	0.122173	0.117684	0.170597	0.153681	0
PSF82	0.120214	0.091032	0.094484	0.162332	0.119272	0.003013
PSF83	0.096706	0.064458	0.089204	0.096769	0.119646	0
PSF84	0.067527	0.062696	0.067333	0.093169	0.080667	0
PSF85	0.086467	0.108251	0.091483	0.157728	0.145104	0.000584
PSF86	0.119113	0.082456	0.116414	0.113026	0.14263	0.010388
PSF87	0.084882	0.069484	0.050577	0.079875	0.11496	0.001431
PSF88	0.107892	0.104053	0.142594	0.102132	0.147791	0.002762
PSF89	0.096265	0.1059	0.115352	0.119085	0.094744	0
PSF90	0.078269	0.057789	0.103817	0.117688	0.117103	0.005363
PSF91	0.088265	0.070851	0.077965	0.098392	0.134328	0
PSF92	0.069034	0.1128	0.122305	0.111774	0.09516	0.003845
PSF93	0.096042	0.066535	0.064253	0.132904	0.104035	0.007474
PSF94	0.104468	0.084374	0.102247	0.141166	0.088541	0.002181
PSF95	0.088082	0.098236	0.0654	0.090577	0.093184	0.00116
PSF96	0.099538	0.1017	0.080208	0.083645	0.100803	0.000702
PSF97	0.146492	0.13896	0.147849	0.135459	0.131312	0
PSF98	0.078135	0.083458	0.073357	0.059454	0.110592	0.008371
PSF99	0.113266	0.118575	0.133691	0.125222	0.13615	0.001849
PSF100	0.014039	0.008307	0.014918	0.014464	0.022042	0.002424
Mean	0.100345	0.099276	0.102893	0.115248	0.130032	0.001425

TABLE S-VIII
COMPARISON OF IGD RESULTS FOR KD-EMMTMA AND EXISTING STATE-OF-THE-ART ALGORITHMS

Instances	IDPSO	TSCOA	KEMMA	MOMA	IMA	KD-EMMTMA
MPSF01	0.015596	0.011648	0.01791	0.018434	0.024389	0.00721
MPSF02	0.025498	0.019224	0.055492	0.043022	0.068567	0.014458
MPSF03	0.048312	0.042496	0.05153	0.040197	0.063732	0.003964
MPSF04	0.092266	0.053333	0.088609	0.084888	0.082109	6.15E-05
MPSF05	0.049878	0.056611	0.042559	0.058879	0.103095	0.00053
MPSF06	0.03777	0.05662	0.054346	0.055352	0.069071	0.001729
MPSF07	0.057268	0.043647	0.066853	0.045707	0.100811	0.008531
MPSF08	0.070906	0.058848	0.083577	0.112049	0.103647	0
MPSF09	0.103676	0.107503	0.108457	0.148523	0.154662	0
MPSF10	0.037182	0.030557	0.03913	0.036236	0.044355	0.003768
PSF01	0.084751	0.077662	0.09918	0.098795	0.1254	0
PSF02	0.080533	0.055794	0.082449	0.082942	0.095652	0.006929
PSF03	0.097241	0.096241	0.093168	0.08331	0.107342	0.000315
PSF04	0.069313	0.067294	0.081543	0.101541	0.123576	0.000778
PSF05	0.110521	0.079293	0.086196	0.085516	0.141828	0
PSF06	0.10902	0.114706	0.120743	0.119531	0.119214	0.003863
PSF07	0.090174	0.091422	0.105526	0.09165	0.107398	0
PSF08	0.058674	0.069448	0.082896	0.08255	0.087936	0
PSF09	0.071114	0.083215	0.076356	0.081522	0.117277	0.004177
PSF10	0.051061	0.093985	0.104356	0.113707	0.119079	0.000788
PSF11	0.101524	0.071372	0.117523	0.119111	0.139744	0.011279
PSF12	0.118869	0.140312	0.107079	0.162631	0.168029	0
PSF13	0.082803	0.066171	0.077939	0.116396	0.110516	0
PSF14	0.090904	0.068465	0.092427	0.127674	0.143249	0.00594
PSF15	0.066218	0.062029	0.08922	0.074033	0.110039	0.008396
PSF16	0.097333	0.08347	0.114138	0.11322	0.119001	0.00635
PSF17	0.105605	0.136628	0.133383	0.130012	0.154987	0.00724
PSF18	0.083669	0.072668	0.09739	0.106855	0.119611	0.003549
PSF19	0.085676	0.049131	0.077949	0.088709	0.115728	0.002821
PSF20	0.0459	0.077313	0.072563	0.067036	0.066909	0.010969
PSF21	0.100338	0.118478	0.101717	0.130865	0.125912	0
PSF22	0.102228	0.079601	0.098275	0.112619	0.103825	0.002524
PSF23	0.120242	0.100741	0.088352	0.106956	0.113599	0.001092
PSF24	0.116869	0.135845	0.118455	0.142777	0.149462	0
PSF25	0.090565	0.081263	0.074712	0.10597	0.105937	0.002185
PSF26	0.100532	0.128638	0.119145	0.106832	0.14189	0.013489
PSF27	0.111717	0.09738	0.083044	0.143	0.110274	0.001125
PSF28	0.074983	0.061363	0.093063	0.09838	0.090101	0.002884
PSF29	0.117716	0.13139	0.082573	0.12682	0.182616	0.011385
PSF30	0.092553	0.103627	0.125853	0.118926	0.101888	0.007181
PSF31	0.066131	0.095601	0.110246	0.132606	0.124982	0.00455
PSF32	0.128942	0.090166	0.111827	0.124941	0.160739	0.004352
PSF33	0.111542	0.146757	0.170051	0.180867	0.220955	0
PSF34	0.109514	0.112063	0.118499	0.105903	0.133579	0.006643
PSF35	0.108259	0.119361	0.077942	0.113144	0.134805	0
PSF36	0.113706	0.131132	0.089403	0.134878	0.115928	0.00084
PSF37	0.108079	0.099174	0.068617	0.13519	0.151065	0
PSF38	0.081253	0.09593	0.110044	0.119964	0.171276	1.59E-05
PSF39	0.114431	0.087718	0.096413	0.139281	0.171177	0.000598
PSF40	0.097987	0.098787	0.085911	0.099383	0.156642	0
PSF41	0.07559	0.081333	0.082917	0.072488	0.108976	0
PSF42	0.110032	0.14031	0.102394	0.155979	0.175246	0
PSF43	0.094267	0.098325	0.113646	0.113568	0.149566	0
PSF44	0.112586	0.098686	0.12159	0.085607	0.108169	0.027984
PSF45	0.090504	0.097267	0.121414	0.15087	0.131172	0
PSF46	0.102044	0.116816	0.129139	0.108898	0.192062	0.000503
PSF47	0.103721	0.085259	0.104036	0.106679	0.157605	0.01232
PSF48	0.091507	0.114984	0.087926	0.147143	0.136714	0
PSF49	0.100643	0.10921	0.071197	0.117314	0.103477	0.00022
PSF50	0.080211	0.123665	0.104439	0.106437	0.123805	0
PSF51	0.146562	0.164164	0.1687	0.144172	0.125705	0
PSF52	0.125806	0.12197	0.125951	0.134935	0.121973	0
PSF53	0.135971	0.118095	0.12129	0.14988	0.098146	0
PSF54	0.100877	0.112223	0.131527	0.124782	0.139292	0.004361
PSF55	0.154174	0.173907	0.126011	0.165897	0.170435	0.010918
PSF56	0.100837	0.087814	0.106825	0.110576	0.110132	0.00402
PSF57	0.086104	0.101903	0.101298	0.118928	0.119167	0
PSF58	0.123854	0.124523	0.113836	0.097773	0.150732	0.000825
PSF59	0.101536	0.122721	0.110877	0.104763	0.145881	0.00209
PSF60	0.062761	0.099589	0.103278	0.07626	0.103708	0.001876
PSF61	0.104769	0.082701	0.115359	0.114727	0.132037	0.005962
PSF62	0.16099	0.098659	0.108935	0.146242	0.141821	0

PSF63	0.128998	0.113702	0.103834	0.125817	0.144229	0.002758
PSF64	0.100508	0.08405	0.100816	0.123252	0.135945	0.001644
PSF65	0.147758	0.142272	0.12491	0.157143	0.149593	0.011626
PSF66	0.142407	0.111924	0.071719	0.118909	0.176372	0.01545
PSF67	0.09658	0.080971	0.097612	0.108611	0.120539	0.003197
PSF68	0.094581	0.086805	0.087069	0.111436	0.104448	0.002644
PSF69	0.126288	0.112877	0.103265	0.10731	0.093492	0.010641
PSF70	0.077256	0.07341	0.074543	0.089618	0.090197	0.001521
PSF71	0.069572	0.084573	0.068063	0.081709	0.08457	0.009263
PSF72	0.126448	0.138636	0.093981	0.116579	0.123932	0.002468
PSF73	0.099053	0.093431	0.081162	0.092828	0.119357	0.002553
PSF74	0.078934	0.078079	0.076987	0.086682	0.08725	0.002934
PSF75	0.066579	0.053268	0.054766	0.080746	0.098307	0.011926
PSF76	0.085727	0.132147	0.100228	0.142958	0.116069	0
PSF77	0.083311	0.138449	0.102419	0.117215	0.163808	0.004412
PSF78	0.126058	0.069673	0.125917	0.131782	0.151051	0.014479
PSF79	0.096488	0.05548	0.118479	0.085155	0.103284	0.005247
PSF80	0.113415	0.094235	0.148449	0.125915	0.158923	0.014085
PSF81	0.105993	0.095095	0.076645	0.122893	0.137264	0.016268
PSF82	0.110394	0.086749	0.091807	0.140718	0.122004	0.003013
PSF83	0.088107	0.071923	0.087593	0.105642	0.116739	0.002793
PSF84	0.065208	0.065822	0.065679	0.088354	0.077284	0.009056
PSF85	0.074054	0.092351	0.08033	0.121768	0.132994	0.000546
PSF86	0.09982	0.08356	0.089211	0.103223	0.120482	0.011442
PSF87	0.082233	0.061797	0.048144	0.071921	0.091255	0.011437
PSF88	0.111109	0.097707	0.123699	0.094421	0.139399	0.002762
PSF89	0.092735	0.10798	0.09426	0.112395	0.107018	0.005936
PSF90	0.079257	0.047412	0.089376	0.107119	0.100783	0.009155
PSF91	0.087481	0.06868	0.082583	0.111859	0.124165	0.016171
PSF92	0.072881	0.09977	0.106765	0.112766	0.076766	0.0024
PSF93	0.093178	0.071101	0.071287	0.111238	0.104556	0.004988
PSF94	0.096021	0.077147	0.098338	0.136162	0.097011	0.003971
PSF95	0.064525	0.075326	0.064551	0.075913	0.073484	0.007085
PSF96	0.077884	0.085734	0.074512	0.072376	0.086593	0.0108
PSF97	0.126194	0.1248	0.135021	0.116365	0.115793	0.002052
PSF98	0.075928	0.071949	0.078007	0.056388	0.096042	0.010978
PSF99	0.1067	0.102729	0.116384	0.111844	0.108767	0.005622
PSF100	0.125754	0.100781	0.074172	0.096927	0.122984	0.00732
Mean	0.094251	0.092351	0.095453	0.108156	0.120838	0.004584

TABLE S-IX
COMPARISON OF HV RESULTS FOR KD-EMMTMA AND EXISTING STATE-OF-THE-ART ALGORITHMS

Instances	IDPSO	TSCOA	KEMMA	MOMA	IMA	KD-EMMTMA
MPSF01	0.891663	0.894175	0.889373	0.889431	0.869154	0.905664
MPSF02	0.820382	0.818675	0.757394	0.777896	0.735424	0.830397
MPSF03	0.788396	0.80157	0.786446	0.81065	0.763324	0.866392
MPSF04	0.705409	0.769282	0.699636	0.718039	0.717864	0.867157
MPSF05	0.904963	0.895682	0.9192	0.878002	0.821169	0.982995
MPSF06	0.872152	0.835894	0.842641	0.846616	0.801395	0.929275
MPSF07	0.845423	0.869036	0.841882	0.864622	0.785701	0.92678
MPSF08	0.859947	0.863061	0.869393	0.79675	0.801685	0.980917
MPSF09	0.777675	0.765498	0.753759	0.689673	0.695007	0.936344
MPSF10	0.745217	0.754314	0.735334	0.742378	0.726271	0.798409
PSF01	0.764349	0.769164	0.740896	0.740668	0.703973	0.890949
PSF02	0.76218	0.823332	0.762696	0.763101	0.732188	0.865848
PSF03	0.79309	0.778201	0.777898	0.786557	0.757654	0.92738
PSF04	0.85164	0.845113	0.829734	0.809316	0.760022	0.958213
PSF05	0.725854	0.763469	0.740446	0.750639	0.668548	0.901892
PSF06	0.79227	0.78757	0.767445	0.774394	0.768508	0.950893
PSF07	0.734856	0.723744	0.698932	0.732531	0.703509	0.875449
PSF08	0.774324	0.770451	0.734589	0.735558	0.738951	0.883455
PSF09	0.724163	0.699956	0.728292	0.707838	0.658181	0.834677
PSF10	0.827677	0.762203	0.748062	0.737137	0.724873	0.915717
PSF11	0.821708	0.865963	0.768748	0.781625	0.732892	0.934225
PSF12	0.774543	0.748154	0.796543	0.707229	0.690974	0.956456
PSF13	0.788211	0.817241	0.78836	0.73117	0.733782	0.901436
PSF14	0.755573	0.789805	0.741836	0.697281	0.674359	0.882798
PSF15	0.914297	0.920862	0.876337	0.90918	0.835721	1.000214
PSF16	0.688739	0.703115	0.652887	0.66114	0.656305	0.838692
PSF17	0.796261	0.753229	0.751934	0.77318	0.715186	0.945798
PSF18	0.758752	0.777412	0.746755	0.744089	0.717498	0.891208
PSF19	0.787588	0.828638	0.771728	0.752662	0.71933	0.905187
PSF20	0.825234	0.757984	0.78804	0.784857	0.778176	0.897388
PSF21	0.736002	0.703405	0.732885	0.687842	0.700577	0.90144
PSF22	0.751533	0.772594	0.754742	0.725998	0.741399	0.9076
PSF23	0.724576	0.75903	0.770377	0.732053	0.740642	0.9122
PSF24	0.757992	0.712245	0.74577	0.728092	0.702347	0.949362
PSF25	0.775909	0.790941	0.784848	0.73327	0.740282	0.911472
PSF26	0.794343	0.742016	0.754107	0.781663	0.741848	0.934861
PSF27	0.785537	0.800569	0.811641	0.736684	0.779171	0.959735
PSF28	0.778275	0.800319	0.763637	0.755951	0.750752	0.892664
PSF29	0.728404	0.689857	0.76922	0.685606	0.635927	0.892053
PSF30	0.728774	0.69636	0.676547	0.68092	0.705234	0.868392
PSF31	0.835546	0.787787	0.763615	0.725951	0.739104	0.925045
PSF32	0.743659	0.790077	0.742247	0.737779	0.663014	0.922355
PSF33	0.801249	0.76578	0.721901	0.691186	0.668893	0.975258
PSF34	0.759298	0.751029	0.754977	0.763867	0.753302	0.908065
PSF35	0.745604	0.71409	0.792003	0.731425	0.690904	0.903281
PSF36	0.711614	0.696303	0.739149	0.694345	0.703696	0.903233
PSF37	0.76818	0.764656	0.808768	0.704381	0.699741	0.928285
PSF38	0.805764	0.765543	0.757721	0.750739	0.678474	0.942018
PSF39	0.776837	0.786508	0.776134	0.729521	0.689	0.945019
PSF40	0.80473	0.776305	0.79107	0.761143	0.70389	0.946357
PSF41	0.798197	0.790307	0.794638	0.7999	0.753917	0.928168
PSF42	0.767804	0.742974	0.807269	0.738479	0.694025	0.954318
PSF43	0.782969	0.781355	0.744203	0.751756	0.716729	0.9395
PSF44	0.765034	0.790924	0.755536	0.799944	0.774708	0.919794
PSF45	0.813763	0.82296	0.799623	0.72742	0.768388	0.966395
PSF46	0.822597	0.804797	0.795921	0.815613	0.714467	0.982769
PSF47	0.76546	0.783123	0.755854	0.76453	0.680479	0.92247
PSF48	0.807658	0.765065	0.795893	0.708103	0.712442	0.939689
PSF49	0.787447	0.801551	0.83786	0.80111	0.805856	0.963762
PSF50	0.760591	0.702878	0.744874	0.718708	0.69179	0.907869
PSF51	0.708131	0.686346	0.681005	0.685912	0.712192	0.919275
PSF52	0.786822	0.782243	0.786131	0.756494	0.771069	0.967883
PSF53	0.785943	0.777383	0.787829	0.74834	0.810608	0.977709
PSF54	0.771648	0.740761	0.712395	0.721828	0.720269	0.9127
PSF55	0.746373	0.731703	0.778773	0.729133	0.724075	0.944409
PSF56	0.714535	0.719313	0.691156	0.687712	0.684522	0.849061
PSF57	0.746259	0.723846	0.715931	0.695809	0.682816	0.888361
PSF58	0.724481	0.726335	0.734934	0.764776	0.690437	0.913215
PSF59	0.756794	0.739648	0.747976	0.760012	0.698124	0.921061
PSF60	0.851944	0.802438	0.795545	0.807924	0.772736	0.943983
PSF61	0.760069	0.789049	0.75189	0.756177	0.725177	0.928819
PSF62	0.695898	0.779447	0.762532	0.70723	0.722137	0.93546

PSF63	0.726658	0.749279	0.760437	0.724469	0.696075	0.922398
PSF64	0.763172	0.763582	0.760622	0.707224	0.689362	0.916365
PSF65	0.70215	0.71363	0.74895	0.686773	0.706672	0.935085
PSF66	0.718732	0.760798	0.82537	0.745382	0.668352	0.940591
PSF67	0.646876	0.662777	0.635592	0.618797	0.607861	0.789148
PSF68	0.693013	0.705604	0.692117	0.653696	0.67646	0.839771
PSF69	0.655287	0.661925	0.677005	0.662779	0.693572	0.840294
PSF70	0.698758	0.688808	0.689602	0.666995	0.672097	0.821138
PSF71	0.678646	0.661307	0.682264	0.64887	0.638246	0.79951
PSF72	0.698929	0.671421	0.735986	0.713624	0.688711	0.886244
PSF73	0.690869	0.691148	0.715753	0.690658	0.656364	0.86473
PSF74	0.72548	0.728071	0.730543	0.718286	0.712158	0.848714
PSF75	0.702208	0.727881	0.719937	0.669454	0.663854	0.827957
PSF76	0.701404	0.651988	0.702734	0.645592	0.683681	0.858699
PSF77	0.840014	0.725828	0.782889	0.759899	0.677498	0.937635
PSF78	0.710103	0.79448	0.71847	0.7264	0.666762	0.910684
PSF79	0.735936	0.775832	0.695715	0.738086	0.700225	0.869228
PSF80	0.729097	0.757601	0.678429	0.711649	0.66281	0.902632
PSF81	0.697245	0.720744	0.740867	0.662536	0.647895	0.869732
PSF82	0.646578	0.688407	0.688745	0.603804	0.647565	0.832392
PSF83	0.664441	0.714379	0.670805	0.654226	0.636469	0.818179
PSF84	0.707423	0.702361	0.705319	0.66368	0.686026	0.802606
PSF85	0.819234	0.788538	0.815414	0.747622	0.737699	0.935023
PSF86	0.698787	0.714675	0.711303	0.690927	0.658002	0.857171
PSF87	0.753878	0.783108	0.804059	0.778634	0.732245	0.881844
PSF88	0.746484	0.769018	0.734183	0.776066	0.69708	0.913449
PSF89	0.708778	0.686687	0.703248	0.661791	0.687424	0.845268
PSF90	0.709844	0.756571	0.700302	0.666746	0.671403	0.823966
PSF91	0.71769	0.740505	0.707736	0.669084	0.646877	0.83047
PSF92	0.73721	0.703707	0.696915	0.691027	0.733488	0.856159
PSF93	0.733356	0.773488	0.782913	0.711673	0.72484	0.870007
PSF94	0.74432	0.782493	0.767527	0.705658	0.755437	0.894562
PSF95	0.725954	0.706019	0.725676	0.702138	0.702256	0.829476
PSF96	0.721767	0.697362	0.718879	0.71506	0.686574	0.847523
PSF97	0.683129	0.707679	0.678897	0.700606	0.69755	0.893225
PSF98	0.705658	0.707963	0.691113	0.74742	0.671538	0.801943
PSF99	0.779524	0.76944	0.780015	0.778088	0.772962	0.940196
PSF100	0.702515	0.736853	0.766023	0.721628	0.70396	0.881402
Mean	0.757527	0.757678	0.753079	0.732878	0.713644	0.900664

K. Application in the Digital Twin System

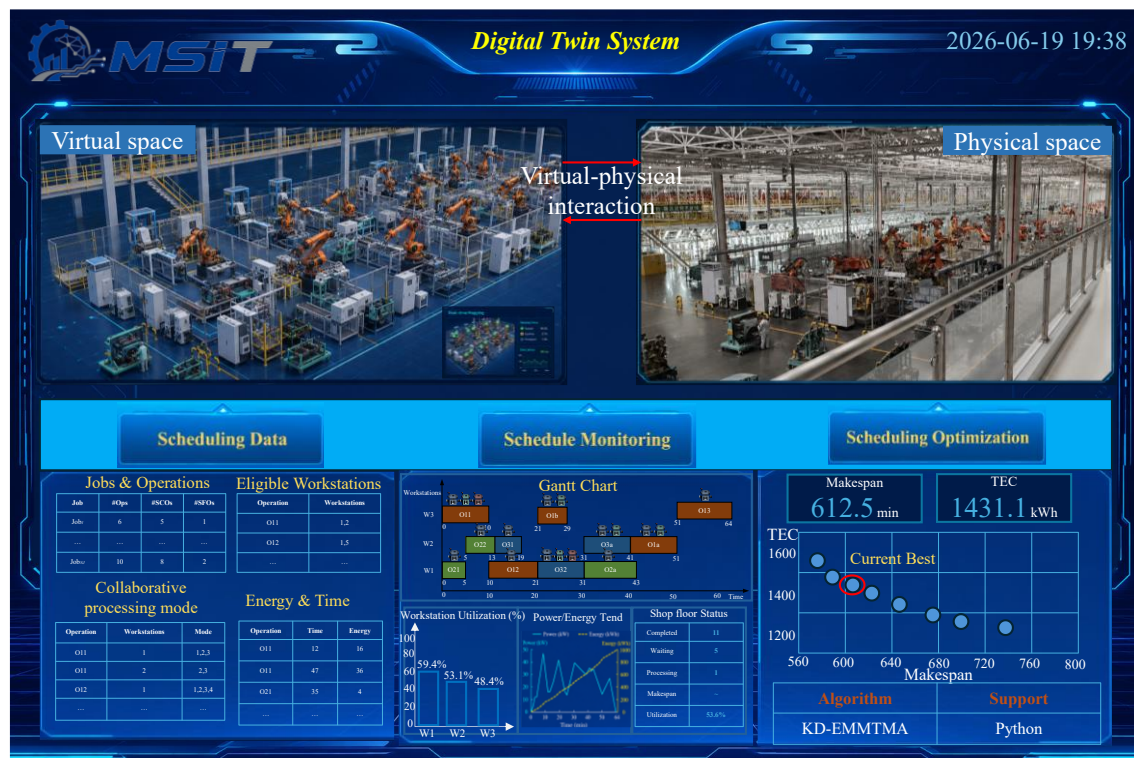


Fig. S-6. Digital twin system for large-flange manufacturing.